

Sharp negative spherical moments when one coefficient is large a partial answer to arXiv:2206.01026

Candidate partial-result packet

13 August 2026

Status: candidate partial result likely valid. The conjectured two-summand constant is proved throughout the source's last negative-moment strip whenever one coefficient carries at least half of the total squared mass. The complementary all-small regime remains open.

1 Source questions

Let $\xi_1, \xi_2, \dots$ be independent random vectors, each uniform on $S^{d-1} \subset \mathbb{R}^d$. For $q > -(d-1)$, the source defines $c_d(q)$ as the best constant such that

$$\left\| \sum_k a_k \xi_k \right\|_q \geq c_d(q) \left(\sum_k a_k^2 \right)^{1/2}. \quad (1)$$

The two-equal-coefficient candidate is

$$c_{d,2}(q) = \left\| \frac{\xi_1 + \xi_2}{\sqrt{2}} \right\|_q.$$

Chasapis, Singh, and Tkocz state that the following ranges appear open and conjecture the two-summand value [1]:

$$c_2(q) = c_{2,2}(q) \quad (-1 < q < 0), \quad c_3(q) = c_{3,2}(q) \quad (-2 < q < -1).$$

very efficient argument, crucially relying on Fourier-analytic formulae for L_p -norms, which together with [39] paved the path for many further results.

That a similar behaviour occurs in the case $d = 2$ (Steinhaus variables) was conjectured by Haagerup, later confirmed by König in [26]: when $d = 2$, $0 \leq q < 2$, we have equality in (4) and the phase transition occurs now at $q_2^* = 0.47\dots$. The range $1 \leq q < 2$ was in fact earlier dealt with by König and Kwapien in [29] (with $q = 1$ handled even earlier by Sawa in [45]), whereas $-1 < q < 0$ (to the best of our knowledge) appears to be left open, with a natural conjecture that $c(q) = c_{2,2}(q)$.

For the case $d = 3$: Latała and Oleszkiewicz showed in [33] that $c_3(q) = c_{3,\infty}(q)$ for $1 \leq q < 2$ which was extended to $0 < q < 1$ in our joint work [9] with Gurushankar (see Proposition 3 below for a connection to uniform distribution on intervals). The phase transition occurs in the range $-1 < q < 0$ at $q_3^* = -0.79\dots$, as established in our joint work [10] with König, so when $d = 3$ and $-1 < q < 2$, (4) holds with equality. Again, $-2 < q < -1$ appears to be open with a natural conjecture that $c(q) = c_{3,2}(q)$.

In higher dimensions $d \geq 4$, there are precise Schur-convexity results available for

The two open ranges in Section 1.1, page 3 of the source PDF.

2 Proof intuition

Write $q = -p$. Because the exponent is negative, the desired lower bound for the L_q norm is an *upper* bound for the singular moment $\mathbb{E}|\sum a_k \xi_k|^{-p}$. Project onto a largest coefficient a and call the remaining random sum Y . Conditional on Y , the moment is a spherical Riesz potential evaluated at $|Y|/a$.

In the last strip $d - 2 < p < d - 1$, the source's hypergeometric expansion has strictly positive nonconstant coefficients. Consequently this potential increases until the two vectors have equal lengths and decreases thereafter. It is therefore bounded by the equal-length value, independently of the random tail. If a contains at least half of the squared mass, the remaining normalization is exactly the conjectured two-summand constant.

3 The partial theorem

For $0 < p < d - 1$, set

$$M_{d,2}(p) = \mathbb{E} \left| \frac{\xi_1 + \xi_2}{\sqrt{2}} \right|^{-p} = c_{d,2}(-p)^{-p}.$$

Theorem 1 (large-coefficient sharp inequality). *Let $d \geq 2$ and $d - 2 < p < d - 1$. For every finite real coefficient vector $a = (a_1, \dots, a_n) \neq 0$ satisfying*

$$\max_k a_k^2 \geq \frac{1}{2} \sum_k a_k^2, \quad (2)$$

one has

$$\mathbb{E} \left| \sum_{k=1}^n a_k \xi_k \right|^{-p} \leq M_{d,2}(p) \left(\sum_{k=1}^n a_k^2 \right)^{-p/2}. \quad (3)$$

Equivalently, (1) holds with the conjectured constant $c_{d,2}(-p)$ for every such vector. Equality in (3) occurs exactly when, up to permutation, signs, and zero coefficients, $|a_1| = |a_2| > 0$.

Thus the theorem covers the whole large-coefficient region for both open low-dimensional ranges, and more generally for the last strip $-(d - 1) < q < -(d - 2)$ in every $d \geq 2$.

4 Proof

Lemma 1 (spherical potential peaks at matching radii). *For $d - 2 < p < d - 1$, let ξ be uniform on S^{d-1} and define*

$$U(r) = \mathbb{E}|re_1 + \xi|^{-p} \quad (r \geq 0).$$

Then $U(r) \leq U(1)$ for every $r \geq 0$, with equality only at $r = 1$.

Proof. Rotational invariance gives $U(r) = \mathbb{E}|e_1 + r\xi|^{-p}$. For $0 \leq r < 1$, the two-coefficient expansion from the source [1, Lemma 13] is

$$U(r) = {}_2F_1 \left(\frac{p}{2}, \frac{p-d+2}{2}; \frac{d}{2}; r^2 \right) = \sum_{m=0}^{\infty} \frac{(p/2)_m ((p-d+2)/2)_m}{(d/2)_m m!} r^{2m}. \quad (4)$$

Both numerator parameters are positive, so every coefficient is positive. The series converges at $r = 1$ because $p < d - 1$; hence U is strictly increasing on $[0, 1]$.

For $r > 1$, homogeneity gives

$$U(r) = r^{-p} U(1/r).$$

With $t = r^{-2}$, the right side is $t^{p/2} F(t)$, where F is the positive-coefficient power series in (4). This expression is strictly increasing in $t \in (0, 1)$, so U is strictly decreasing in $r \in (1, \infty)$. The unique maximum is therefore at $r = 1$. $\square$

Proof of Theorem 1. By symmetry take $a = |a_1| = \max_k |a_k|$ and put $Y = \sum_{k=2}^n a_k \xi_k$. Condition on Y . Averaging over ξ_1 and rotating the fixed vector Y to the first coordinate gives, by Lemma 1,

$$\begin{aligned}\mathbb{E} |a\xi_1 + Y|^{-p} &= a^{-p} \mathbb{E} U(|Y|/a) \\ &\leq a^{-p} U(1).\end{aligned}$$

The law of a pair of independent spherical vectors and rotational invariance give

$$M_{d,2}(p) = 2^{p/2} \mathbb{E} |\xi_1 + \xi_2|^{-p} = 2^{p/2} U(1).$$

Writing $A^2 = \sum_k a_k^2$, condition (2) now yields

$$a^{-p} U(1) \leq 2^{p/2} A^{-p} U(1) = M_{d,2}(p) A^{-p},$$

which proves (3). Taking both sides to the power $-1/p$ gives the equivalent L_{-p} inequality.

For equality, both steps must be equalities. Thus $a^2 = A^2/2$ and $|Y| = a$ almost surely. If the nonzero tail coefficients are b_j , then independence and $\mathbb{E} \langle \xi_i, \xi_j \rangle^2 = 1/d$ give

$$\text{Var } |Y|^2 = \frac{4}{d} \sum_{i < j} b_i^2 b_j^2.$$

Constancy of $|Y|$ forces at most one nonzero tail coefficient. Its squared magnitude is $A^2 - a^2 = a^2$, proving the equality assertion. Conversely, two coefficients of equal magnitude plainly give equality. $\square$

5 Upgrade audit: the all-small obstruction

The theorem does not cover $\max a_k^2 < A^2/2$. Two materially different upgrade routes were pursued.

Three summands in $d = 3$. When a, b, c are all small, they obey the triangle inequalities. Put $\beta = 3 - p \in (1, 2)$ and write $a = y + z$, $b = x + z$, $c = x + y$. Explicit spherical convolution reduces the full three-summand claim, after $x + y + z = 1$, to

$$(1 - x^\beta - y^\beta - z^\beta)(1 + x^2 + y^2 + z^2)^{(3-\beta)/2} \leq \beta 2^{(3-\beta)/2} (1-x)(1-y)(1-z). \quad (5)$$

The endpoint $\beta = 2$ follows from $xyz \leq (xy + yz + zx)^2/3$; the endpoint $\beta = 1$ is trivial. Finite-grid checks found no interior violation, but no valid interpolation for the coupled exponents in (5) was obtained. It is therefore recorded only as an unproved route, not as part of the result.

Arbitrary all-small vectors. The source's Fourier/Hölder method reduces the missing estimates to sharp weighted Ball-type inequalities

$$s^{p/2} \int_0^\infty |j_{d/2-1}(t)|^s t^{p-1} dt \leq 2^{p/2} \int_0^\infty |j_{d/2-1}(t)|^2 t^{p-1} dt, \quad s \geq 2.$$

Here $j_0 = J_0$ for $d = 2$ and $j_{1/2}(t) = \sin(t)/t$ for $d = 3$. The integrals converge in exactly the open strips, but equality at $s = 2$ makes nonsharp Gaussian or tail bounds insufficient. This is the remaining sharp analytic barrier.

6 Verification, novelty status, and limitations

The proof was checked at the exponent endpoints, under scaling, and in the equality case. The identity for $\text{Var } |Y|^2$ was independently expanded term by term. The script in `code/check_upgrade_reduction.py` checks the unproved three-summand reduction on finite grids; it is not used as proof.

A bounded search on 13 August 2026 covered the run indexes, arXiv:2206.01026, the exact open-range sentences, and combinations of “negative spherical Khinchin”, “largest coefficient”, “half the squared mass”, and “spherical Riesz potential”. No explicit statement of this partial theorem or later full resolution was found. Novelty confidence is moderate: the key hypergeometric expansion is in the source and the projection argument is short, but this application to the two stated open ranges appears unrecorded.

The result is a coefficient-region theorem, not a proof of either complete conjecture. Human review should focus on the reciprocity step in Lemma 1, conditioning at a possibly zero tail, and the equality case. The recommended disposition is to retain this as a substantial partial result and pursue (5) or the weighted Bessel inequality separately.

References

- [1] G. Chasapis, S. Singh, and T. Tkocz, *Haagerup’s phase transition at polydisc slicing*, arXiv:2206.01026 (2022; version dated 2023). <https://arxiv.org/abs/2206.01026>.